

Effect of Metasurface Quality Factor on the Thermal Time Constant in Microbolometers

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Abstract— This paper investigates the effects of the metasurface quality on the thermal mass of metasurface integrated microbolometers. In this configuration, a temperature sensitive layer forms the insulator in a metal/insulator/metal metasurface. The geometric parameters of a hexagonal close packed disk array are optimized for perfect absorption at $\lambda=10.6\ \mu\text{m}$ for different thicknesses. Increasing the thickness of the insulator increases the quality factor but increases the thermal mass of the microbolometer.

Keywords—metasurface, microbolometer, Q factor, thermal response time.

I. INTRODUCTION

Microbolometers are a common uncooled infrared detector. They are less expensive and more portable than cooled sensors [1]. Microbolometers are heated by absorbing thermal radiation. The increased temperature changes the resistance sensitivity layer's resistance, which can be detected as output voltage or current signals. Traditional microbolometers are designed to achieve broadband absorption in order to maximize the response from thermal radiation. This is typically accomplished using a quarter wavelength cavity structure. The support legs of the pixel must be placed outside of the cavity which reduces the fill-factor when the microbolometers are arranged in a focal plane array. Using a metasurface to directly absorb infrared radiation removes this requirement and provides a path toward spectral selectivity. These metasurfaces typically include a ground plane and the temperature sensitive resistance layer must be electrically isolated. While the metasurface can be placed above the temperature sensitive layer, serving as a paint, this significantly increases the thermal mass of the microbolometer. Integrating the temperature sensitive resistance layer as part of the metasurface insulating layer improves this but the thickness of the stack must still be selected to achieve the desired absorption [2]. Either configuration allows the supporting legs to be moved under the pixel increase the fill factor. Spectroscopic applications are also potentially enabled by the metasurface. The absorption peak can be tuned by modifying metasurface geometry without changing the thickness of the insulator [3], which simplifies the fabrication of multi-microbolometer arrays. The most common temperature sensitive materials in microbolometers are based on VO_x or amorphous silicon. While VO_x has a less low frequency noise

[4], amorphous has much higher resistivity and lower imaginary permittivity [5].

In this work, a simple metasurface with an integrated amorphous silicon layer is designed. This is intended to model a metasurface integrated microbolometer. The metasurface is optimized to be a perfect absorber at $10.6\ \mu\text{m}$. The results illustrate that the quality factor scales with the thermal response time which has design considerations for future devices.

II. METASURFACE INTEGRATED MICROBOLOMETER

A. Structure of Metasurface Integrated Microbolometer

Figure 1 shows a microbolometer integrated with a metasurface. The bottom ground plane is 50 nm gold and top is 25 nm thickness hexagonal close packed (HCP) array of disks. An amorphous silicon layer with thickness $t_{a\text{-Si}}$, lies immediately underneath the disks. This layer is insulated from the ground plane by a 50 nm thick ZnSe layer.

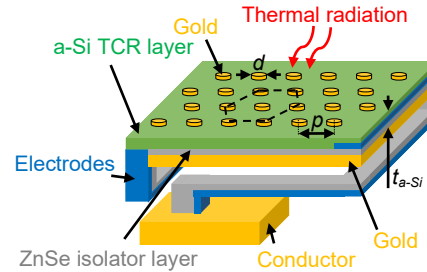


Fig. 1. Diagram of metasurface integrated microbolometer.

While SiO_2 and Si_3N_4 are more likely to be used in standard CMOS fabrication [6,7], they have a more complicated permittivity. We have selected ZnSe to illustrate the effects of a comparably lower index dielectric. The absorptance of the HCP disk array is relatively insensitive to the polarization and angle of incidence of the incident radiation. The energy absorbed by metasurface heats amorphous silicon layer. The increased temperature changes amorphous silicon layer's resistance which can be measured using the supporting electrodes.

B. Thermal Mass

The thermal response time of this metasurface integrated microbolometer can be calculated as

$$\tau = \frac{C}{G} \quad (1)$$

where G is the thermal conductance of legs and C is the thermal mass. The thermal conductance depends on the geometry and material of the support structure with $G=1 \times 10^{-7}$ W/K assumed for this paper. The thermal mass is given by

$$C = c_{\text{gold}} \rho_{\text{gold}} (V_{\text{gold-disk}} + V_{\text{gold-ground}}) + c_{\text{a-Si}} \rho_{\text{a-Si}} V_{\text{a-Si}} + c_{\text{ZnSe}} \rho_{\text{ZnSe}} V_{\text{ZnSe}} \quad (2)$$

where c is the specific heat capacity, ρ is the density and V is the volume of the microbolometer. A pixel size of $40 \times 40 \mu\text{m}^2$ is assumed in this paper, although this can also be calculated on a per area basis, $C''=C/A$. The volume of gold disk averaged over the area. The average absorption of this metasurface integrated microbolometer can be defined as

$$\bar{\alpha} = \frac{1}{6} \int_8^{14} \alpha(\lambda) d\lambda \quad (3)$$

where 8- 14 μm is the long wavelength infrared range.

C. Absorptance for Perfect Metasurface Absorber

Figure 2 shows the infrared absorbance spectrum of metasurface integrated microbolometer calculated by HFSS with the permittivities of gold, a-Si and ZnSe taken from [8-10]. These perfect absorbers at 10.6 μm are optimized by fixing $t_{\text{a-Si}}$ and using an optimization routine to find p and d . The optimized geometry parameters and total thermal mass are listed in Table I.

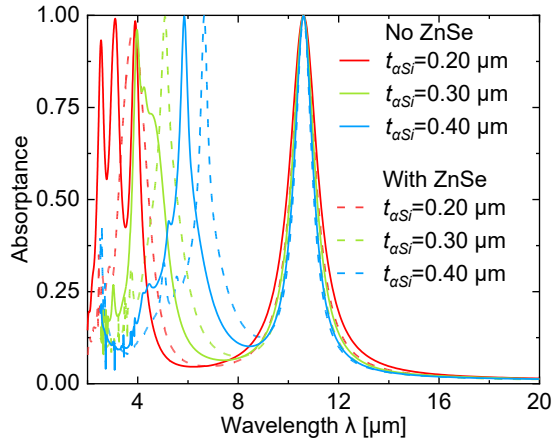


Fig. 2. Optimized metasurface integrated microbolometer absorption.

Narrower absorption resonant peaks are obtained when the a-Si layers are thicker. Capacitance between adjacent disks compensates a thinner a-Si layer. Adding ZnSe adds thermal mass but reduces the effective permittivity of the combined insulating layer. For both cases, the periodicity, p , is proportional with a-Si thickness $t_{\text{a-Si}}$. The disk diameter, d , is dependent on the wavelength and is not strongly dependent on the thickness, although the reduced effective index of refraction results in larger disks for the thinner layers. Increasing $t_{\text{a-Si}}$, thermal mass also increased. For the modest range of quality

factors, Q , the average absorption over the LWIR range does not change significantly.

D. Q Factor

Figure 3 shows the relationship between the quality factor and area normalized thermal mass, C'' . The addition of ZnSe lowers the thermal mass for the same Q . This tradeoff has implications for the spectral selectivity and response time or sampling rate when the metasurface is used by itself to determine the bandwidth. Better results may be obtained by comparing the signals from multiple microbolometers.

Table I. The parameters for metasurface perfect absorber at $\lambda = 10.6 \mu\text{m}$

a-Si thickness ($t_{\text{a-Si}}$) [μm]	ZnSe thickness [μm]	Diameter (d) [μm]	Period (p) [μm]	Thermal mass [J/K]	Average absorption
0.15	0	1.34	1.61	6.51×10^{-10}	0.30
0.2	0	1.40	2.10	7.61×10^{-10}	0.27
0.25	0	1.41	2.63	8.80×10^{-10}	0.26
0.3	0	1.41	3.22	1.00×10^{-9}	0.24
0.35	0	1.39	3.78	1.13×10^{-9}	0.23
0.4	0	1.36	4.48	1.26×10^{-9}	0.23
0.15	0.05	1.56	2.45	7.69×10^{-10}	0.34
0.2	0.05	1.54	3.00	8.88×10^{-10}	0.31
0.25	0.05	1.50	3.52	1.01×10^{-9}	0.29
0.3	0.05	1.47	4.16	1.14×10^{-9}	0.26
0.35	0.05	1.43	4.78	1.27×10^{-9}	0.25
0.4	0.05	1.38	5.46	1.40×10^{-9}	0.24

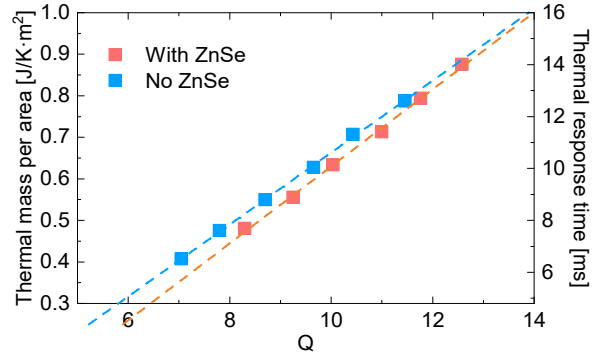


Fig. 3. Comparison of Q factor between metasurfaces with and without ZnSe layer.

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